Association Rule Learning

DSC 340 Applied Machine Learning

EMC Ch. 5

April 14, 2025

The Plan

- Presently studying unsupervised learning (no labels!)
  - Clustering (k-Means)
  - Dimensionality Reduction (PCA)
  - Time Series (ARIMA)
- Today's topic: Association Rules (Apriori)
  - What is it? [EMC 5.1]
  - Applications [5.4]
  - Apriori algorithm [5.2]
  - Evaluating results [5.3, 5.6]
  - Helpful package [apyori]

Overview

EMC 5.1

Association Rule Learning

- **Unsupervised**
- Data mining problem in business (dates to 1960s)
- Input: **Transactions** in a database
  - Example: shopping cart (**"market basket analysis"**)
- Output: **Rules**
  - Apocryphal example: {diapers} → {beer}
- Evaluation: Should be **"interesting" and "useful"**
  - Business context: should lead to insights that can be operationalized
  - For example, provide new insights into buyer behavior

Rules

- Form: $A \rightarrow B$
  - When A is observed, B is also observed (i.e., in the cart)
  - "Correlation does not imply causation" … but in business we may not care!
- Discovery of interesting patterns
  - Note: could be used in data exploration phase of any analytics project
- Suppose we sell N items in our grocery
  - How many possible subsets are there?
- Problem: Number of rules is exponential in the number of products!
- Strategy: Prune rules that lack "support"

Support

- Threshold for inclusion in set of rules
  - AR: support for **itemset** **not class** (in classification)
- Percentage of transactions that contain an itemset *L*
  - Example: if 60% of all transactions contain {bread, milk} then **support** of {bread, milk} is 0.6.
- Use as criterion: minimum threshold
- Question: What are the **parameters?**
  - The rules!
- What are the **hyperparameters?**
  - Minimum threshold! Also max length of itemset. Number of iterations …

Applications

EMC 5.4

Applications of ARs

- Better merchandizing
  - What products should be included in my inventory each month?
- Cross-merchandizing
- Physical or logical placement of product
  - Grocer could include display of diapers at the end of the beer aisle
  - Or could include a coupon for beer in the diaper aisle
  - Online shopping: "Often bought together" (Amazon)
- Promotional programs
  - loyalty card incentives, coupons mailed to customers, etc.
- Many others (recommender systems, even database sequencing!)

Apriori Algorithm

EMC 5.2

Apriori algorithm

- Agrawal and Srikant (1994)
- Bottom-up: Starts with 1-itemsets, then 2-itemsets, etc.
- Extended one at a time
- Candidates discovered each iteration
- Candidate subset added if it meets threshold
- Downward closure property
- Uses breadth-first search and Hash tree for efficient operation

$\text{Apriori}(T, \epsilon)$

$L_1 \leftarrow \{\text{large } 1-\text{itemsets}\}$

$k \leftarrow 2$

while $L_{k-1} \neq \emptyset$

$C_k \leftarrow \{c = a \cup \{b\} \mid a \in L_{k-1} \wedge b \notin a, \{s \subseteq c \mid |s| = k-1\} \subseteq L_{k-1}\}$

for transactions $t \in T$

$D_t \leftarrow \{c \in C_k \mid c \subseteq t\}$

for candidates $c \in D_t$

$count[c] \leftarrow count[c] + 1$

$L_k \leftarrow \{c \in C_k \mid count[c] \geq \epsilon\}$

$k \leftarrow k + 1$

return $\bigcup_k L_k$

https://en.wikipedia.org/wiki/Apriori_algorithm

Evaluation

EMC 5.3, 5.6

Evaluation

- **Confidence**: measure of certainty or trustworthiness of discovery

$$Confidence\ (X \rightarrow Y) = \frac{Support(X \wedge Y)}{Support(X)}$$

- **Lift:** how much more often X and Y occur together than expected

$$Lift(X \rightarrow Y) = \frac{Support(X \wedge Y)}{Support(X) * Support(Y)}$$

- **Leverage:** same idea, but uses **difference** rather than **ratio**

$$Leverage\ (X \rightarrow Y) = Support(X \wedge Y)\ - Support(X) * Support(Y)$$

Apriori at a Glance

- Set a minimum value for **support** and **confidence**
- **Extract subsets** having **support** > minimum threshold
- **Select** all **rules** from **subsets** having **confidence** > minimum threshold
- **Order** the rules by descending order of **lift**

Package

https://pypi.org/project/apyori/

Questions?

What are your questions?